Joint Distributions

Bivariate distribution

Definition

X, Y: discrete random variables defined on the same sample space.

A, B: sets of possible values of X and Y.

$$p(x, y) = P(X = x, Y = y) = P(A \cap B)$$

$$\sum_{x \in A} \sum_{y \in B} p(x, y) = 1$$

Probability mass function of X

$$p(x) = P(X = x) = P(x = x, y \in B) = \sum_{y \in B} p(x, y)$$

Theorem

p(x, y): Joint probability mass function of discrete R.V. X and Y.

h(X, Y): A function of X and Y

$$E[h(X, Y)] = \sum_{x \in A} \sum_{y \in B} h(x, y) p(x, y)$$

$$E(X + Y) = E(X) + E(Y)$$

Proof

$$\begin{aligned} E[X+Y] &= \sum_{x \in A} \sum_{y \in B} (x+y) p(x, y) \\ &= \sum_{x \in A} \sum_{y \in B} x p(x, y) + \sum_{x \in A} \sum_{y \in B} y p(x, y) \\ &= \sum_{x \in A} x (\sum_{y \in B} p(x, y)) + \sum_{y \in B} y (\sum_{x \in A} p(x, y)) \\ &= \sum_{x \in A} x p_X(x) + \sum_{y \in B} y p_Y(y) = E[X] + E[Y] \end{aligned}$$

Definition

Two r. v.'s X and Y, defined on the same sample space, have a continuous joint distribution if there exists a nonnegative function of 2 variables, f(x,y) on $R \times R$ such that for some region S in the xy-plane

$$P((X, Y) \in S) = \iint_S f(x, y) dx dy$$

$f(x, y)$: is called the joint (probability) density function of X and Y.

Definition

單點機率為 0 : $P(X=a, Y=b)=\int_b^b\left(\int_a^a f(x,y)dx\right)dy=0$

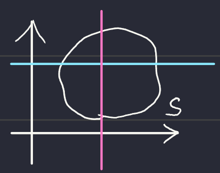
總和機率為 1 : $A=(-\infty,+\infty), B=(-\infty,+\infty) \implies \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)dxdy=1$

Definition

Let X and Y have joint density function f(x,y)

$f_Y(y)=\int_{-\infty}^{\infty} f(x,y)dx$

$f_X(x)=\int_{-\infty}^{\infty} f(x,y)dy$



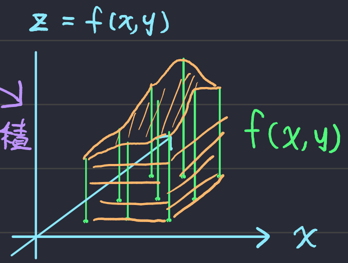
are called, the marginal (probability) density functions of X and Y.

Definition

Joint (cumulative) distribution function

$$F(X,Y)=F(t,u)=P(X\le t, Y\le u)=\int_{-\infty}^{y}\int_{-\infty}^{x} f(u,t)dydu$$

求體積



The marginal distribution function

$F_X(t)=P(X\le t)=P(X\le t, Y\le\infty)=F(t,\infty)$

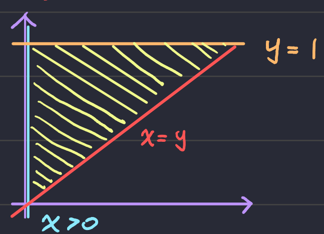
$F_Y(t)=P(Y\le t)=P(X\le\infty, Y\le t)=F(\infty,t)$

Ex 8.4 The joint density function of random variables X and Y is given by

$$f(x,y)=\begin{cases}\lambda xy^2 & 0\le x\le y\le 1\\ 0 & \text{otherwise.}\end{cases}$$

(a) Determine the value of lambda

繪圖找範圍



$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$

$=\int_0^1\int_0^y \lambda xy^2\,dx\,dy$

$=\int_0^1 \lambda\left(\frac{1}{2}x^2y^2\right)\Big|_0^y dy$

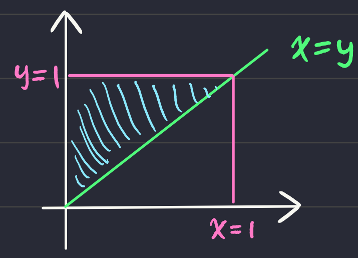
$=\lambda\int_0^1 \frac{1}{2}y^4\,dy$

$=\lambda\left(\frac{1}{10}y^5\right)\Big|_0^1$

$=\frac{1}{10}\lambda$

Ans : $\lambda = 10$ #

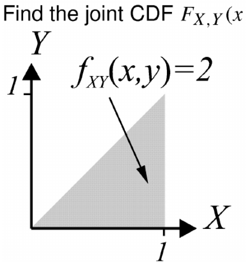
(b) Find the marginal density function of X and Y.　作答注意範圍



$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$
$$= \int_x^1 f(x,y)\,dy$$
$$= 10\left(\tfrac{1}{3}\,x\,y^3\right)\Big|_x^1$$
$$= \tfrac{10}{3}\left(x - x^4\right) \;\#\qquad \text{if } 0 \le x \le 1$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$
$$= \int_0^y 10\,x\,y^2\,dx$$
$$= 5\,x^2 y^2 \Big|_0^y$$
$$= 5\,y^4 \;\#\qquad \text{if } 0 \le y \le 1$$

Find the joint CDF $F_{X,Y}(x,y)$ when X and Y have joint PDF



$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \le y \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

$$F(t,u) = P(X \le t,\ Y \le u)$$
$$= \int_{-\infty}^{Y}\int_{-\infty}^{X} f(x,y)\,dt\,du$$
$$= \int_0^Y \int_u^X 2\,dt\,du$$
$$= \int_0^Y (2t)\Big|_u^X\,du$$
$$= \int_0^Y 2X - 2u\,du$$
$$= (2Xu - u^2)\Big|_0^Y$$
$$= 2XY - Y^2$$

後面還有一坨範圍判定

Ex 8.5

For $\lambda > 0$, let

$$F(x, y) = \begin{cases} 1 - \lambda e^{-\lambda(x+y)} & \text{if } x > 0,\ y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Determine if F is the joint distribution function of 2 random variables X and Y.

$$\frac{\partial^2}{\partial x \partial y} F(x, y)$$

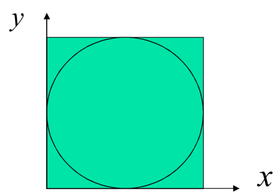
$$= \frac{\partial}{\partial x} -\lambda e^{-\lambda(x+y)} \cdot (-\lambda)$$

$$= \frac{\partial}{\partial x} \lambda^2 e^{-\lambda(x+y)}$$

$$= -\lambda^3 e^{-\lambda(x+y)} \quad \text{if } x>0,\ y>0$$

↳ 負的，所以不行 #

Ex 8.6 **A circle of radius 1 is inscribed in a square with sides of length 2. A point is selected at random from the square. What is the probability that the point is inside the circle?**



$\frac{\pi}{4}$ #

$$f(x,y) = \begin{cases} 1 & x^2+y^2 < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 8.2

$$E(Z) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y)f(x,y)dxdy$$

$Z = h(x,y)$

$f(x,y)$: The probability density function

Corollary

For continuous r. v.'s X and Y, $E(X+Y) = E(X) + E(Y)$

$$\begin{aligned} E(X+Y) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x+y)f(x,y)dxdy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dxdy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y)dxdy \\ &= E(X) + E(Y). \end{aligned}$$

marginal density function

Ex 8.9 **Let X and Y have joint density function**

$$f(x,y) = \begin{cases} \frac{3}{2}(x^2+y^2) & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X^2+Y^2)$

Let $h(x,y) = x^2 + y^2$

$$\begin{aligned} E(X^2+Y^2) &= E(h(x,y)) \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y) \cdot f(x,y)\,dx\,dy \\ &= \int_0^1\int_0^1 h(x,y) \cdot \tfrac{3}{2} h(x,y)\,dx\,dy \\ &= \tfrac{3}{2} \cdots \end{aligned}$$

8.2 Independent random variables

Definition

A, B are called independent if $P(\in A, Y \in B) = P(X \in A)P(Y \in B)$

If A, B are independent, then :

Probability $P(X \leq a, Y \leq b) = P(X \leq a)P(Y \leq b)$

Joint distrubution function $F(t, u) = P(x < t)P(y < u) = F_x(t)F_y(u)$

Joint PMF $p(x, y) = p_x(x)p_y(y)$ 離散

Joint density function $f(x, y) = f_x(x)f_y(y)$ 連續

Theorem 8.5

X, Y: two independent R.V.

g, h: two R $\longrightarrow$ R function

$\longrightarrow$ g(x) and h(Y) are also independent

Theorem 8.6

X, Y: two independent R.V

g, h: two R $\longrightarrow$ R function

$E[g(X)h(Y)] = E[g(x)]E[h(Y)]$

本理論不是雙向的，如果

$E[g(X)h(Y)] = E[g(x)]E[h(Y)]$

成立，不代表他們一定獨立

$\boxed{\text{Proof}}$ $E(Z) = E[g(x)h(y)]$

$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} [g(x)h(y)] \cdot f(x,y)\, dx\, dy$

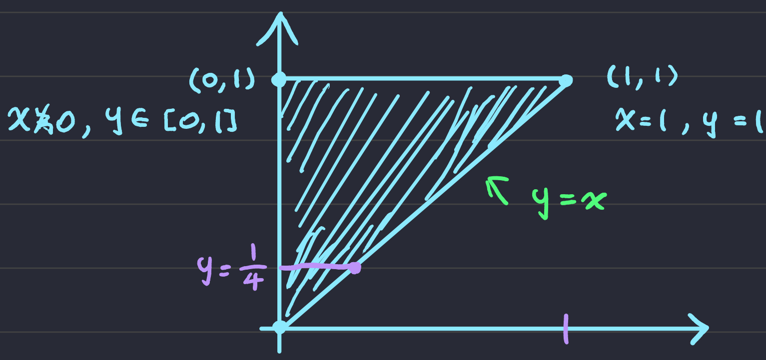
$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} [g(x)h(y)] \cdot [f_X(x) \cdot f_Y(y)]\, dx dy$

$= \int_{-\infty}^{\infty} h(y) f_Y(y)\, dy \cdot \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$

$= E[h(y)] \cdot E[g(x)]$

Ex 8.15 **Prove that 2 random variables X and Y with the following joint density function are not independent.**

$$f(x,y) = \begin{cases} 8xy & 0 \le x \le y \le 1 \\ 0 & \text{otherwise.} \end{cases}$$



找 x 範圍

1. 找隨意 y 值
2. 看 x 的 Range
3. 0 ~ y

8.3 Conditional distributions

$$p_{X|Y}(x|y) = p_X(x|y) = \frac{p(x,y)}{p(y)}$$
$$= p(x|y)$$

All Be Okay

Ex 8.21 **Let X and Y be continuous random variables with joint density function**

$$f(x,y) = \begin{cases} \frac{3}{2}(x^2+y^2) & \text{if } 0<x<1,\ 0<y<1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $f_{X|Y}(x|y)$.

$$f(x|y) = \frac{f(x,y)}{f(y)}$$
$$f(y) = \int_0^1 \frac{3}{2}(x^2+y^2)\,dx$$
$$= \frac{3}{2}y^2 + \left(\frac{1}{2}x^3\right)\Big|_0^1$$
$$= \frac{3}{2}y^2 + \frac{1}{2}$$

$$f_{X|Y}(x|y) = \frac{3/2(x^2+y^2)}{(3/2)y^2+1/2} = \frac{3(x^2+y^2)}{3y^2+1}$$

for $0<x<1$ and $0<y<1$.

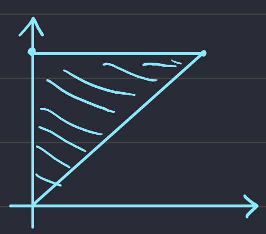
Everywhere else, $f_{X|Y}(x|y) = 0$.

Ex 8.22 **First, a point Y is selected at random from the interval (0,1).Then another point X is chosen from the interval (0,Y). Find the density function of X.**

$$f(Y) = \begin{cases} 1 & \text{if } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x|Y) = \begin{cases} \frac{1}{y} & \text{if } 0 \le x \le y,\ 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x|Y) = \frac{f(x,y)}{f(Y)} \qquad f(x,y) = f(x|Y)\cdot f(Y)$$
$$= \frac{1}{y} \quad \text{if } 0 \le x \le y \le 1$$



$$f(x) = \int_x^1 \frac{1}{y}\,dy$$
$$= -\ln(x) \quad \text{if } 0 \le x \le 1,\ \text{elsewhere } 0 \ \#$$

Ex 8.24 **Let X and Y be continuous random variables with joint density function**

$$f(x, y) = \begin{cases} e^{-y} & \text{if } y > 0,\ 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X \mid Y = 2)$.

$E(X \mid Y = 2)$

$= \int_{-\infty}^{\infty} x \cdot f(x,2)\, dx$

$f(x,2) = \dfrac{f(x,2)}{f_y(2)}$

$f_y(2) = \int_{-\infty}^{\infty} f(x,2)\, dx$

$= \int_0^1 e^{-2}\, dx = e^{-2}$